A Delay Differential Equation Model for Tumor Growth

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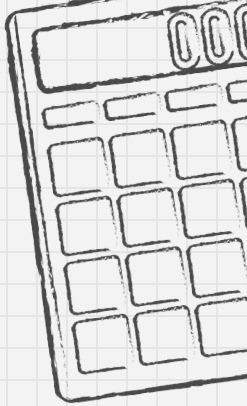
Agenda for today



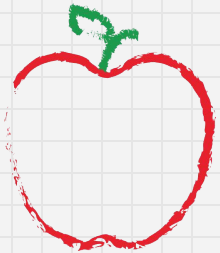
01
Introduction
Foundational knowledge base

02
Mathematical Model
Differential description of the model

03
Model Analysis
Summary of the analysis done in the paper

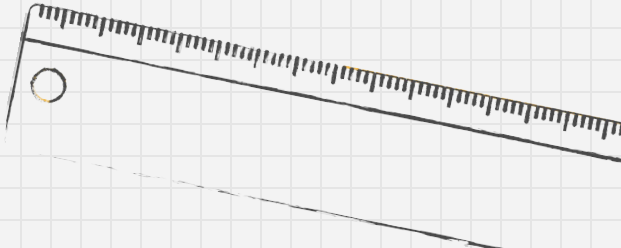
04
Model Extension
Cancer Cells in their Quiescent Phase

01



Introduction

Cancer statistics and the motivation behind the project

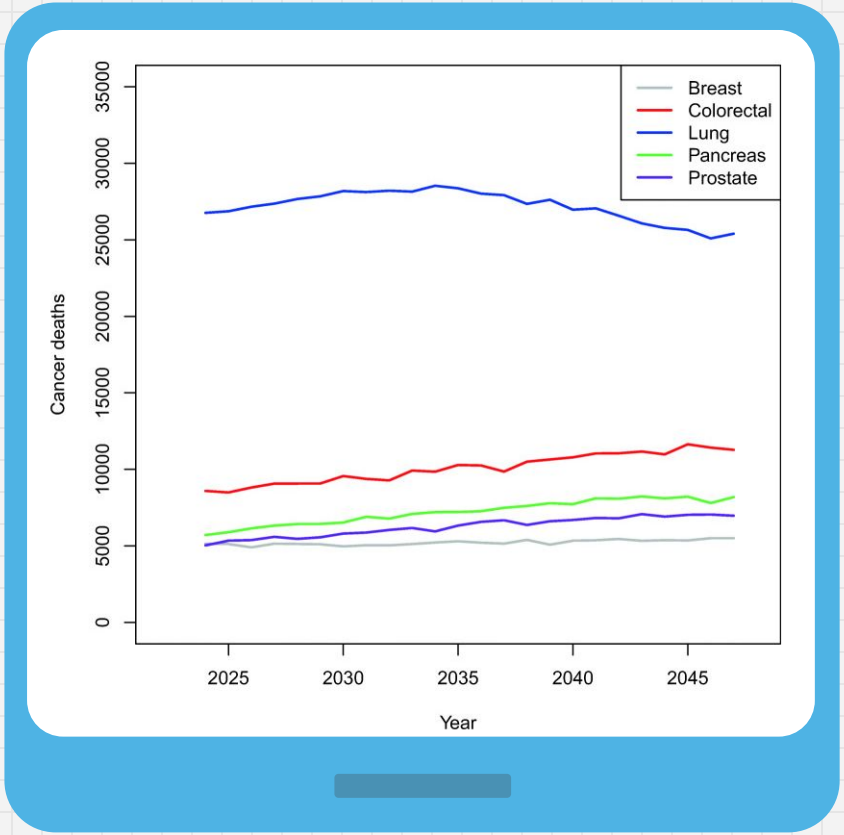


Cancer in Canada , 2024-2047

89,850 → 112,920

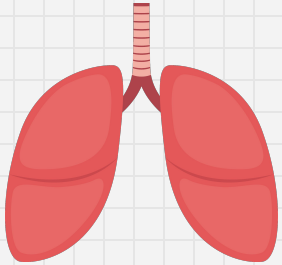
2024 2047

Projected cancer death toll rise

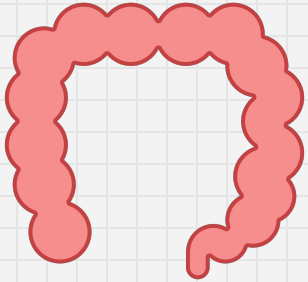


Source: Pader, Joy et al. "Estimates of future cancer mortality attributable to modifiable risk factors in Canada." Canadian journal of public health vol. 112,6 (2021): 1069-1082. doi:10.17269/s41997-020-00455-7

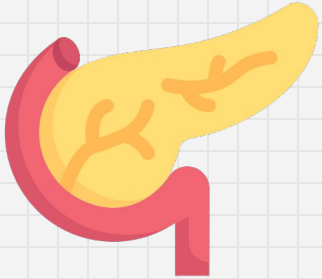
Categorized Projections (2047)



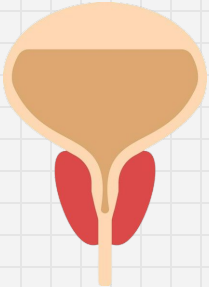
Lung
25, 400



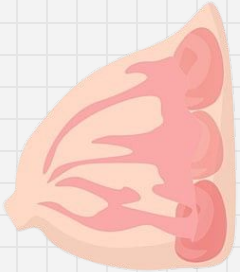
Colorectal
11, 280



Pancreas
8,190



Prostate
6,970



Breast
5,510

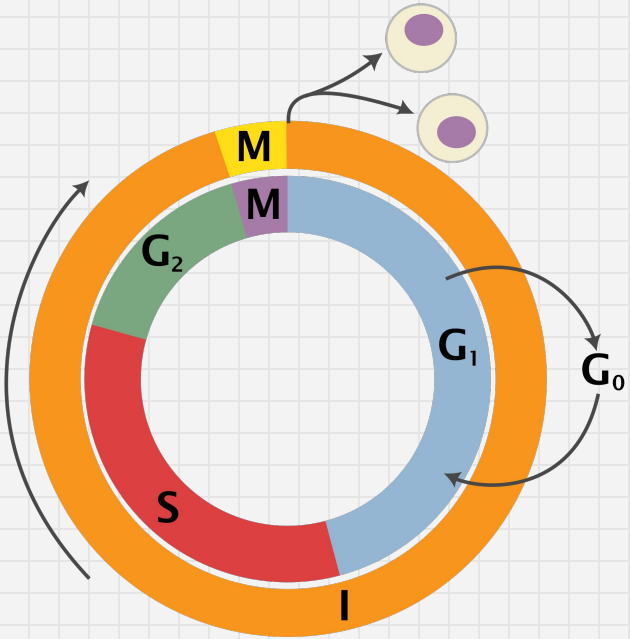
Source: Pader, Joy et al. “Estimates of future cancer mortality attributable to modifiable risk factors in Canada.” Canadian journal of public health vol. 112,6 (2021): 1069-1082. doi:10.17269/s41997-020-00455-7

The Cell Cycle

The Cell Cycle is is the series of events that take place in a cell that causes it to divide into two daughter cells. It can be divided into four phases:

- *G1* - gap period
- *S* - synthetic period
- *G2* - post-synthetic gap period
- *M* - mitosis

The most typical normal cell will have a cell cycle duration of *~24* hours.

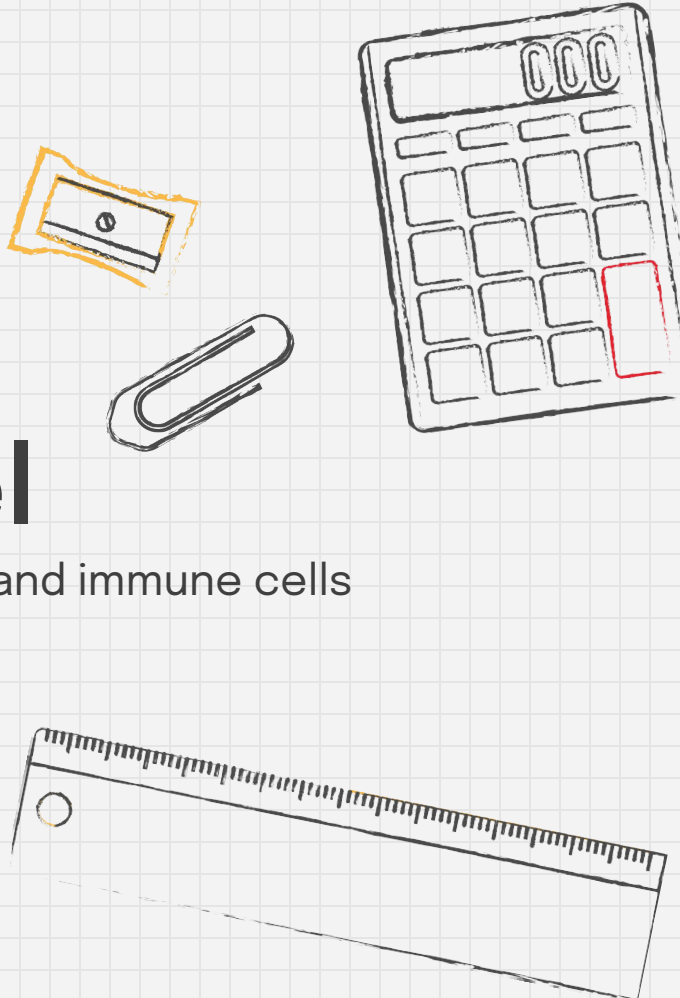


Media source: Richard Wheeler (2006). Schematic representation of the cell cycle. Cytokinesis forms rapidly in the process of the cell cycle. Wikimedia Commons.

02

Mathematical Model

The effects and interactions between tumor cells and immune cells



Main Research Focus

A delay differential equation model for tumor growth

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PMID: 12955460 DOI: 10.1007/s00285-003-0211-0

Species of Interest

T_I

Tumor cells
(Interphase)

T_M

Tumor cells
(mitosis)

I

Immune cells

u

Drugs

Assumptions & Simplifications

- Mitosis is an instantaneous process.
- The drug is cytotoxic.
- The drug only arrests tumor cells in mitosis.
- The Quiescent period G_0 is excluded.
- The concentration of the drug decays exponentially with time.

Parameter Naming Conventions

a terms	Rates of reproduction
c terms	Tumor-immune combat loss
ϱ, α, n	Depends on the **type of tumor** being considered and the **health of the immune system**
τ	Interphase period

Species of Interest

T_I

Tumor cells
(Interphase)

T_M

Tumor cells
(mitosis)

I

Immune cells

u

Drugs

Dynamics - T_I

Interphase tumor cells

$$T_I' = \overbrace{2a_4 T_M}^{\text{Cells entering interphase after mitosis}} - \underbrace{(c_1 I}_{\text{Loss from immune cells}} + \overbrace{d_2}^{\text{Apoptosis}})T_I - \underbrace{a_1 T_I(t-\tau)}_{\text{Interphase cells entering mitosis}}$$

Dynamics - T_M

Mitosis tumor cells

$$T'_M = \overbrace{a_1 T_I(t-\tau)}^{\text{Interphase cells entering mitosis}} - \underbrace{d_3 T_M}_{\text{Apoptosis}} - \overbrace{a_4 T_M}^{\text{Cells entering interphase after mitosis}} - \underbrace{c_3 T_M I}_{\text{Loss from immune cells}} - \overbrace{k_1(1-e^{-k_2 u})T_M}^{\text{Drug interaction}}$$

Dynamics - T_I and T_M

Interphase and mitosis tumor cells

$$T_I' = 2a_4 T_M - (c_1 I + d_2) T_I - a_1 T_I(t-\tau)$$

$$T_M' = a_1 T_I(t-\tau) - d_3 T_M - a_4 T_M - c_3 T_M I - k_1(1 - e^{-k_2 u}) T_M$$

$$T_I \rightleftarrows T_M$$

Tumor cells (Interphase)

Tumor cells (mitosis)

Dynamics - I

Immune cells

$$I' = \overbrace{k}^{\text{Idle growth rate}} + \underbrace{\frac{\rho I (T_I + T_M)^n}{\alpha + (T_I + T_M)^n}}_{\text{Growth due to stimulus (Michaelis-Menten)}} - \overbrace{c_2 I T_I}^{\text{Combat loss}} - \underbrace{c_4 T_M I}_{\text{Combat loss}} - \overbrace{d_1 I}^{\text{Apoptosis}} - \underbrace{k_3 (1 - e^{-k_4 u}) I}_{\text{Drug interaction}}$$

The drug

affects both

mitosis tumor cells and immune cells

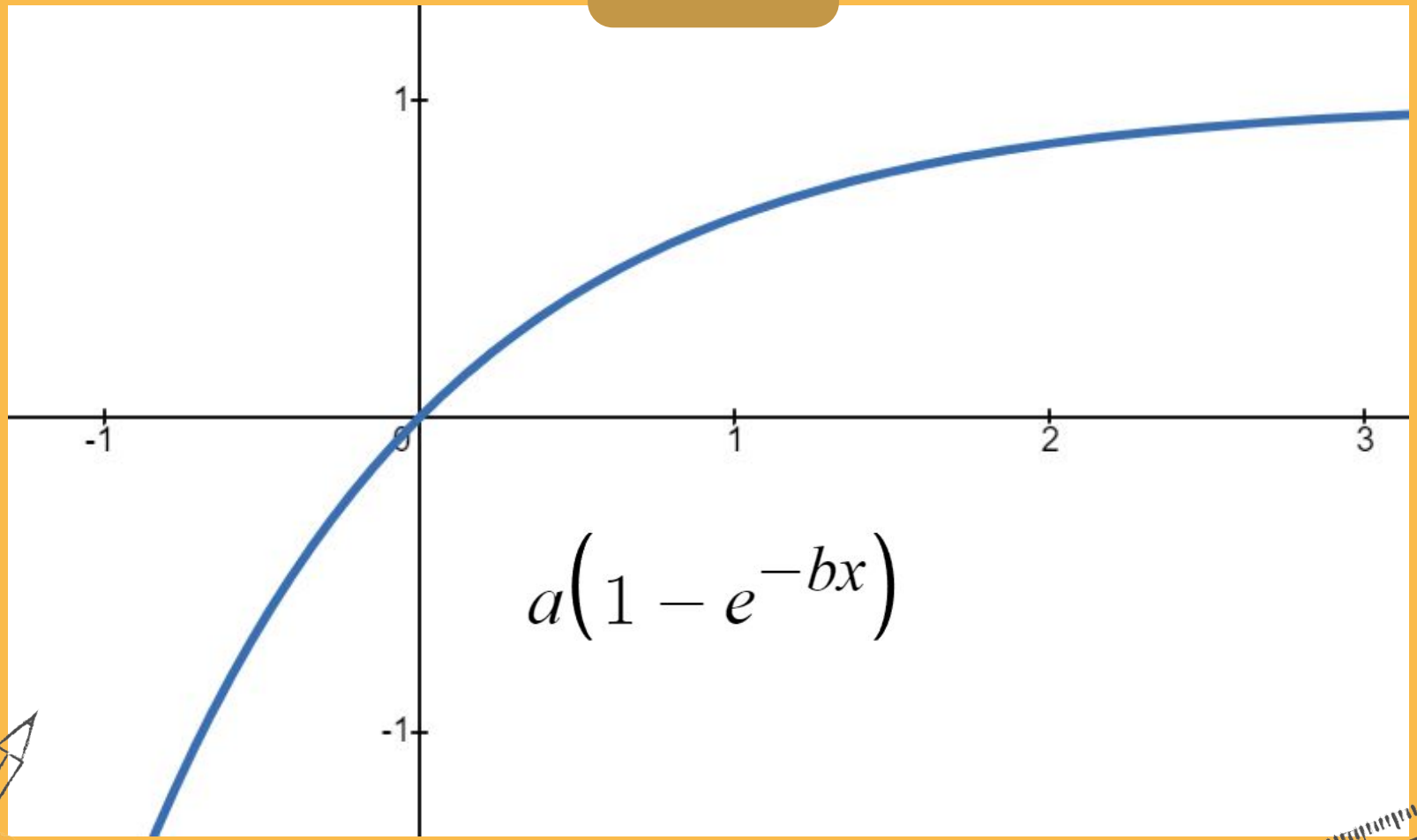
but not

interphase tumor cells

$$T_I' = 2a_4 T_M - (c_1 I + d_2) T_I - a_1 T_I(t-\tau)$$

$$T_M' = a_1 T_I(t-\tau) - d_3 T_M - a_4 T_M - c_3 T_M I - k_1(1 - e^{-k_2 u}) T_M$$

$$I' = k + \frac{\rho I (T_I + T_M)^n}{\alpha + (T_I + T_M)^n} - c_2 I T_I - c_4 T_M I - d_1 I - k_3(1 - e^{-k_4 u}) I$$



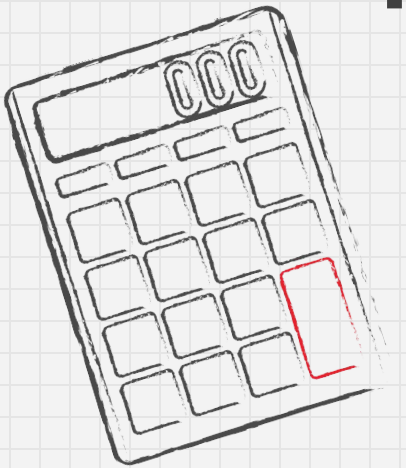
$$a\left(1 - e^{-bx}\right)$$

Dynamics - u

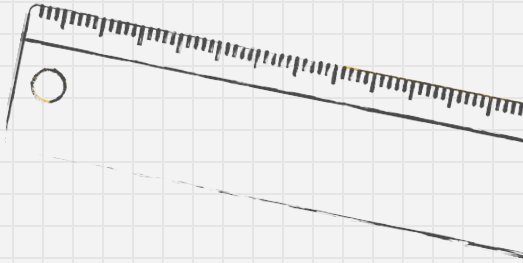
Drug concentration

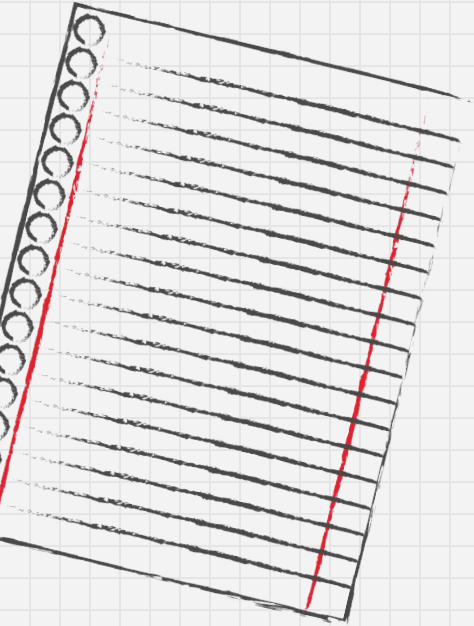
$$u' = -\gamma u$$

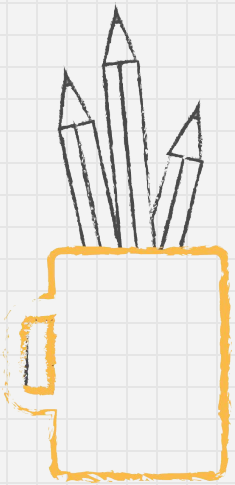
Parameter Values



Parameter	Estimated value
τ	22 hr. (0.9167 days)
a_1	0.8470 day^{-1}
a_4	0.9159 day^{-1}
d_2	0.1145 day^{-1}
d_3	0.6641 day^{-1}
$c_1 = c_3$	2.16×10^{-7} cell^{-1} day^{-1}
$c_2 = c_4$	3.422×10^{-10} cell^{-1} day^{-1} day^{-1}
k	1.3×10^4cell day^{-1}
d_1	0.04 day^{-1}
n	3
ρ	0.2 day^{-1}
α	$(0.3 \times 10^6\text{cell})^3$







03

Model Analysis

Analysis of the model

Drug-free,
Non-delay,
No immune suppression

$$
\begin{aligned}
T_I' &= 2a_4 T_M - (d_2 + a_1) T_I, \\
T_M' &= a_1 T_I - \underbrace{d}_{d \;=\; d_3 + a_4} T_M,
\end{aligned}
$$

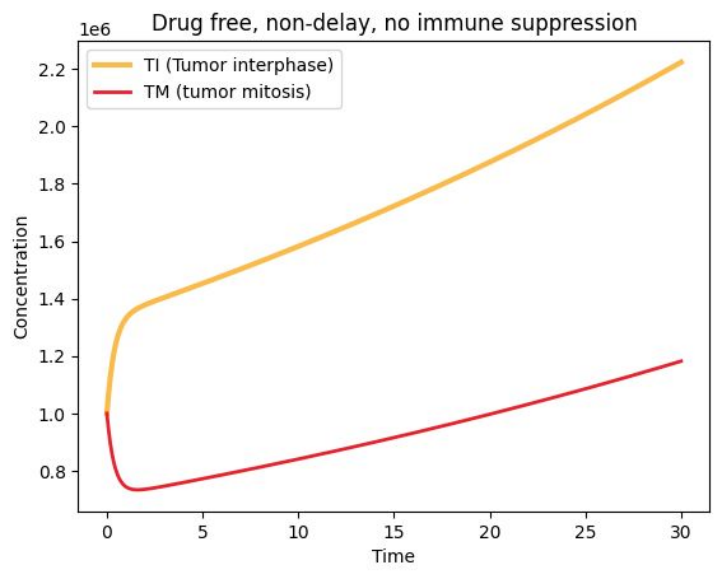
Drug-free,
Non-delay,
No immune suppression

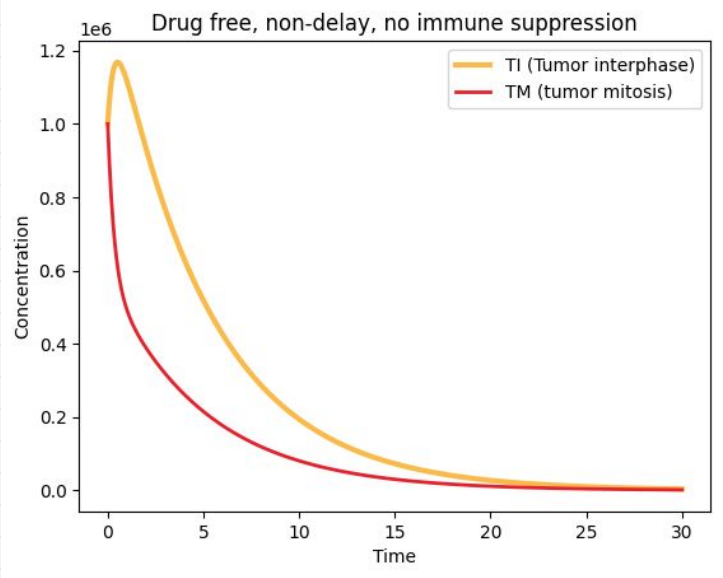
$$d(d_2 + a_1) < 2a_1a_4$$

Death **Growth**

$$d(d_2 + a_1) > 2a_1a_4$$

Death Growth





Drug-free,
Non-delay,
No immune suppression

Drug-free,
Non-delay,
No immune suppression

$$T_I' = 2a_4 T_M - d_2 T_I - a_1 T_I(t-\tau),$$
$$T_M' = a_1 T_I(t-\tau) - d T_M,$$

Drug-free,
No immune suppression

Stability of (0,0)

$$x = \frac{\left(-(d_2^2 + d^2 - a_1^2) \pm \sqrt{(d_2^2 + d^2 - a_1^2)^2 - 4d_2^2 d^2 + 4a_1^2 (d - 2a_4)^2}\right)}{2}$$

Unstable
(Tumor grows)

$$\Delta < 0$$

Stability switch
(Tumor grows if tau large)

$$\begin{cases} d > 2a_4 \\ d(a_1 - d_2) > 2a_1 a_4 \\ \Delta > 0 \end{cases}$$

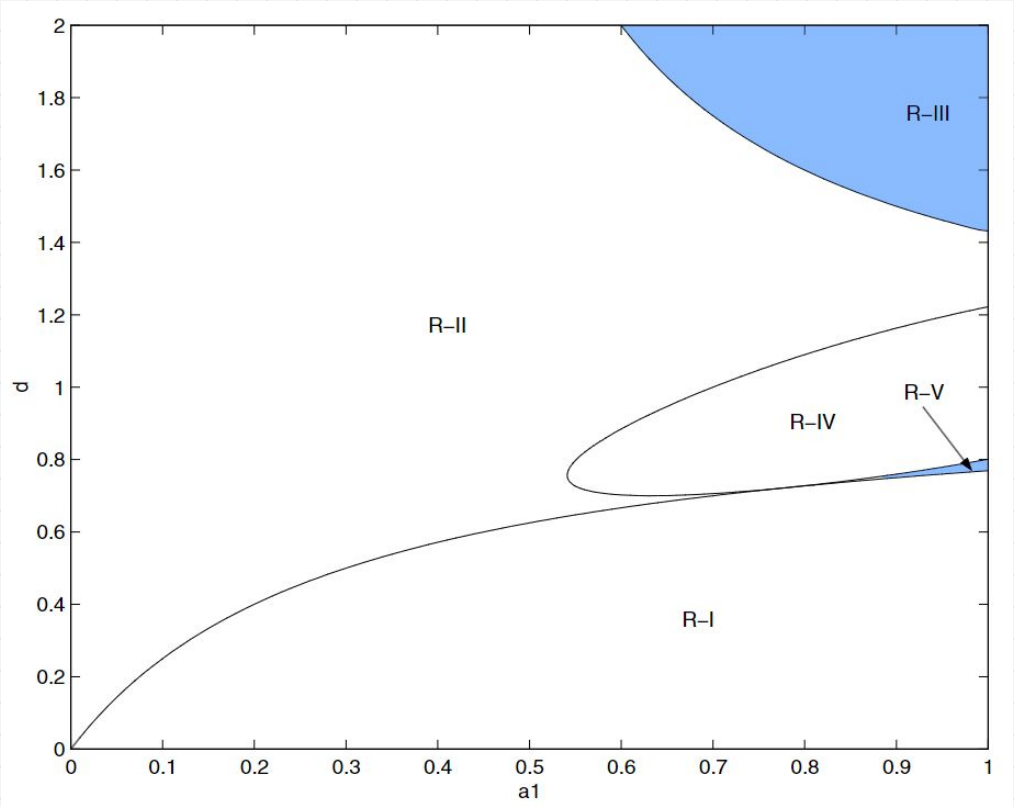
$$\begin{cases} \text{the quantity inside the radical is positive} \\ d_2^2 + d^2 - a_1^2 < 0 \\ \Delta > 0 \end{cases}$$

Stable
(Tumor shrinks)

All other cases

Drug-free,
No immune suppression

Stability of (0,0)



Drug-free,
No immune suppression

Drug-free,
Non-delay,
No immune suppression

$$
\begin{aligned}
F(\lambda) &= P(\lambda) + e^{-\tau\lambda} Q(\lambda) \\
&= \lambda^3 + p_2\lambda^2 + p_1\lambda + p_0 + e^{(-\tau\lambda)} \left[q_2\lambda^2 + q_1\lambda + q_0 \right]
\end{aligned}
$$

Drug-free

Theorem 1. *Let* $P(\lambda) = (\lambda+\beta_1)(\lambda+\beta_2)(\lambda+\beta_3)$ *and* $Q(\lambda) = (\lambda+\alpha_1)(\lambda+\alpha_2)$, *such that*

$$d := \prod_{i=1}^{3} \beta_i^2 - \prod_{i=1}^{2} \alpha_i^2 > 0, \quad \prod_{i=1}^{3} \beta_i > 0, \quad \prod_{i=1}^{2} \alpha_i < 0,$$

$$\sum 1/\beta_j > \sum 1/\alpha_k, \quad \text{and} \quad \sum \alpha_j > 0.$$

Assume that (4) is stable for $\tau = 0$.

Let $b := \sum \beta_i^2 - 1$, $c := \sum_{i \neq j} \beta_i^2 \beta_j^2 - \sum \alpha_i^2$ *and* $R := (-b + \sqrt{b^2 - 3c})/3$. *Then if:*

1. $c < 0$ *and if* $R(-b^2 + 6c + b\sqrt{b^2 - 3c}) + 9d < 0$, *then there exists a* $\tau^* > 0$, *such that for* $\tau > \tau^*$ *are at least two roots of (4) on the right hand side of the complex plane.*
2. $c > 0$ *and* $b > 0$, *then there are no roots on the right hand side of the complex plane.*
3. $c > 0$, $b < 0$, $b^2 - 3c > 0$ *and* $R(-b^2 + 6c + b\sqrt{b^2 - 3c}) + 9d < 0$, *then there exists a* $\tau^* > 0$, *such that for* $\tau > \tau^*$ *there are at least two roots of (4) on the right hand side of the complex plane.*

Drug-free

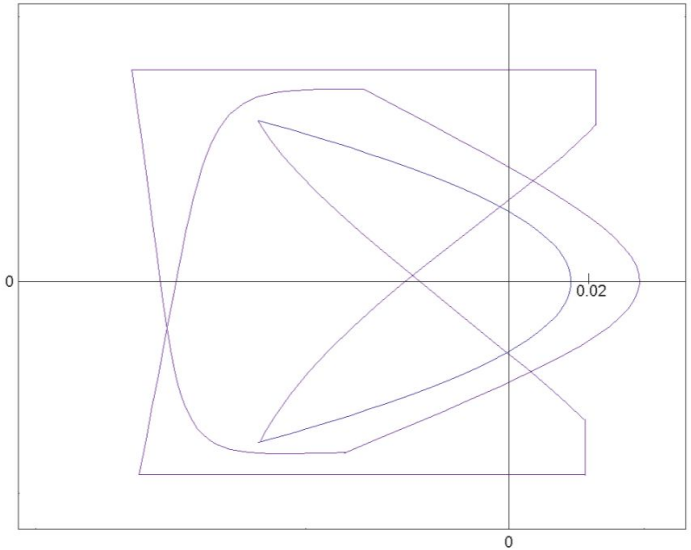


Fig. 4. $F \circ \gamma$ for $\tau = 1$. The parameter vales have been set to yield a stable fixed point, and this is seen as $F \circ \gamma$ gives a net of 0 encirclements as we traverse the contour γ.

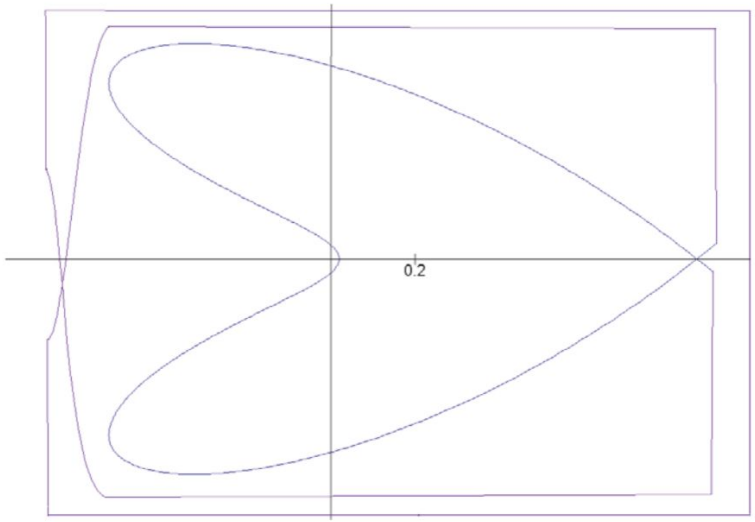


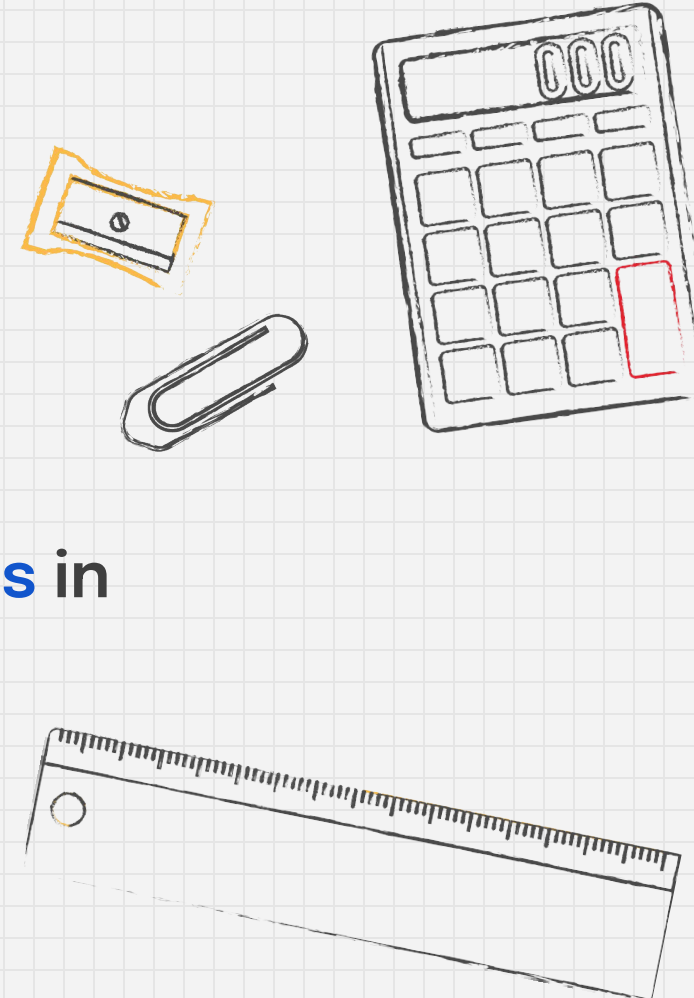
Fig. 5. $F \circ \gamma$ for $\tau = 6$. As we increase the delay τ, the fixed point changed its stability from being stable (as in the case $\tau = 1$) to unstable. Theorem 1 predicts this change of stability as there is a net of 2 encirclements as we traverse the contour.

Drug-free

04

Model Extension

Consideration of the Cancer Cells in their Quiescent Phase G_0

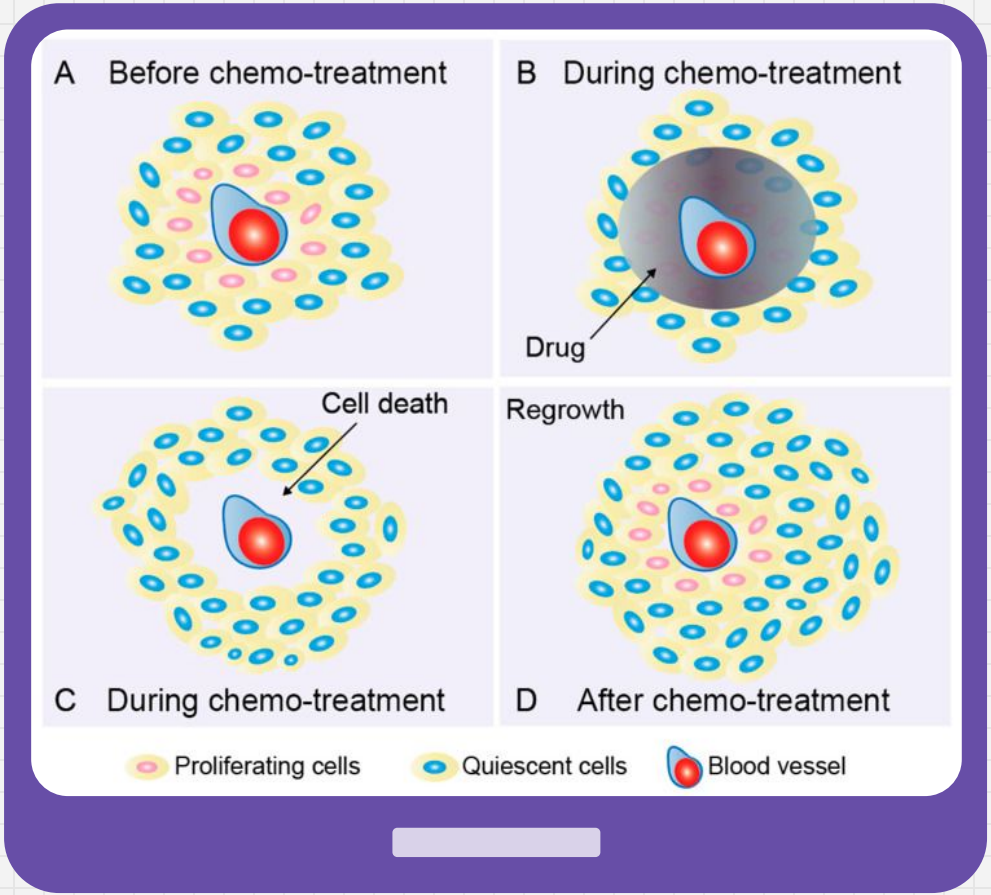


Quiescent Phase G_0

What is a Quiescent Cancer Cell (QCCs)?

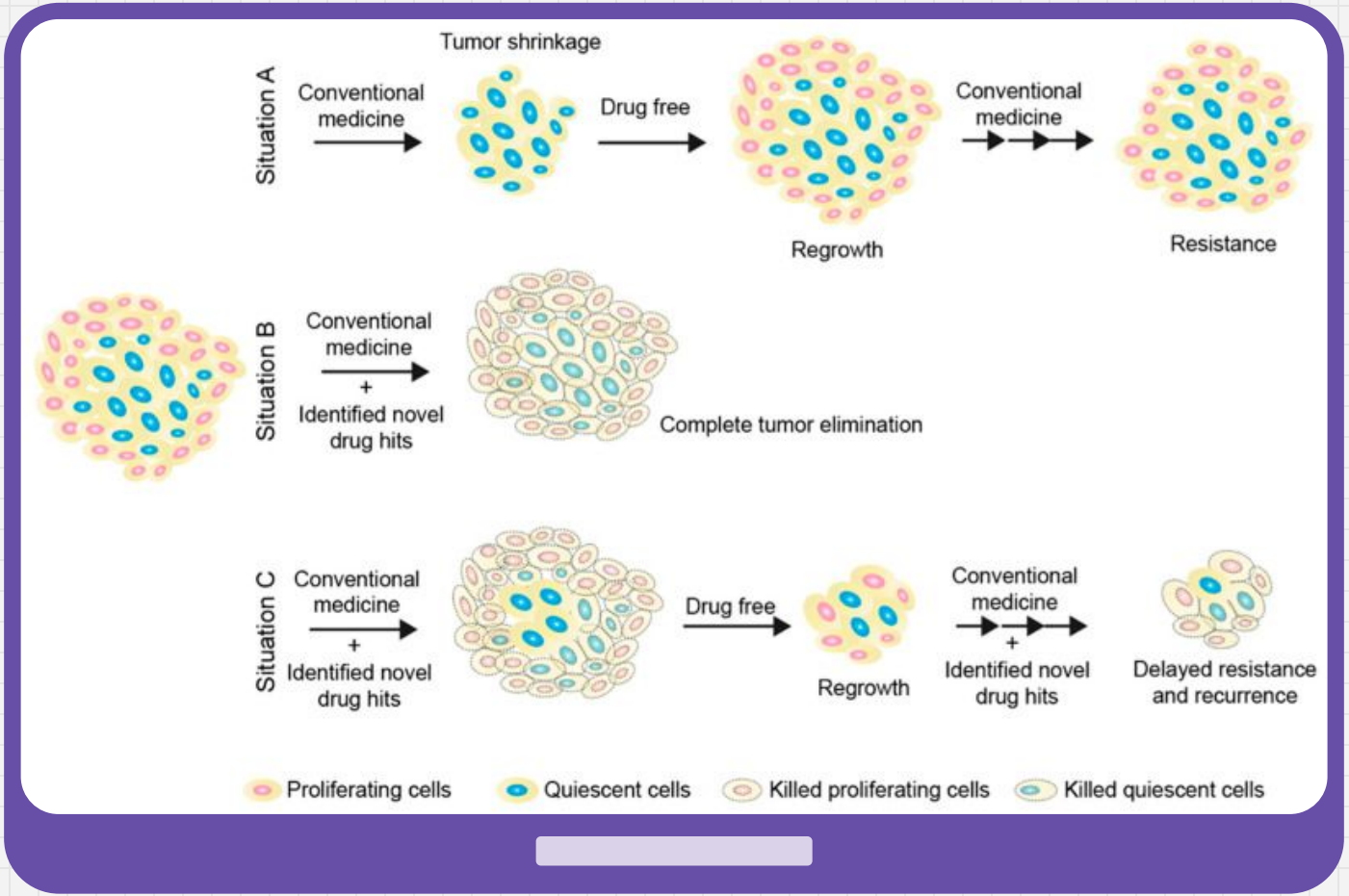
They are cancer cells that are reversibly suspended in the G_0 phase, with the ability to re-enter the cell cycle and initiate cancerous tumor growth.

QCCs lead to drug resistance and tumor recurrence



Source: Lindell, E., Zhong, L., & Zhang, X. (2023). Quiescent Cancer Cells-A Potential Therapeutic Target to Overcome Tumor Resistance and Relapse. International journal of molecular sciences, 24(4), 3762. https://doi.org/10.3390/ijms24043762

Elimination of tumor cells by targeting both proliferative and quiescent cancer cell populations



Source: Lindell, E., Zhong, L., & Zhang, X. (2023). Quiescent Cancer Cells-A Potential Therapeutic Target to Overcome Tumor Resistance and Relapse. International journal of molecular sciences, 24(4), 3762. https://doi.org/10.3390/ijms24043762

New Species of Interest

T_Q

Tumor cells
(Quiescent)

New Parameters

a_q	Rate of production of QCCs
d_q	Rate of transformation of QCCs to Proliferative Cancer Cells
τ_q	Time taken for QCCs to exit quiescent phase

New Dynamics - T_Q

Quiescent Cancer Cells

$$T_q' = \overbrace{a_q T_M}^{\text{Cells entering } G_0} - \underbrace{d_q T_Q(t-\tau)}_{\text{Cells exiting } G_0}$$

Modified Dynamics - T_I

Interphase tumor cells

$$T_I' = 2a_4 T_M - (c_1 I + d_2) T_I - a_1 T_I(t-\tau) + d_q T_Q(t-\tau)$$

Modified Dynamics - T_M

Mitosis tumor cells

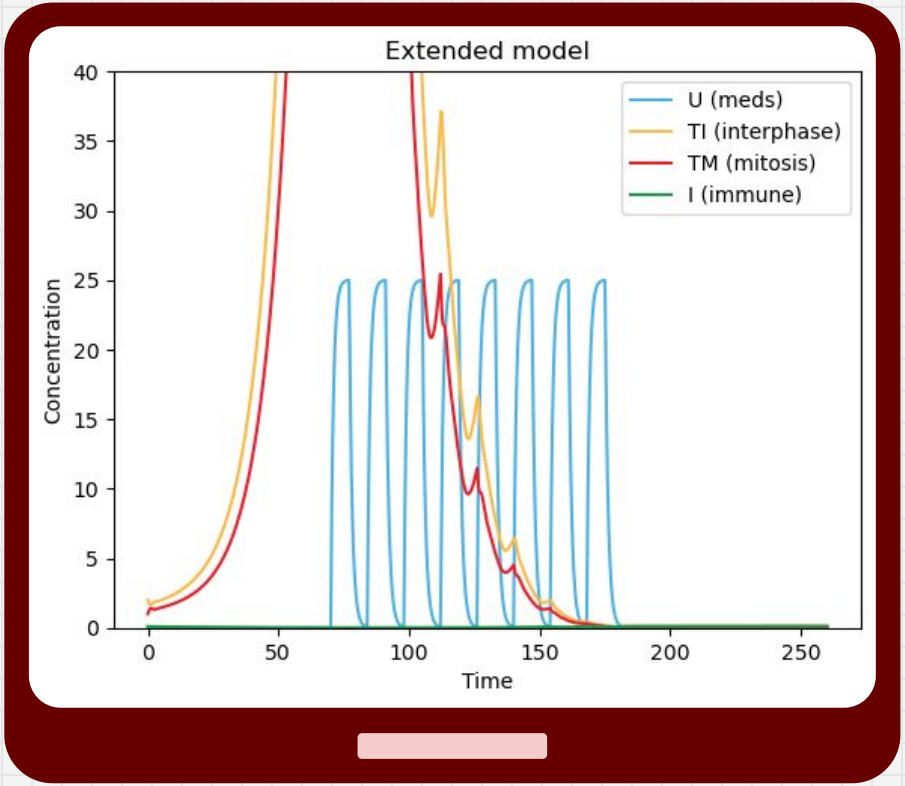
$$T'_M = a_1 T_I(t-\tau) - d_3 T_M - a_4 T_M - c_3 T_M I - k_1(1-e^{-k_2 u})T_M$$

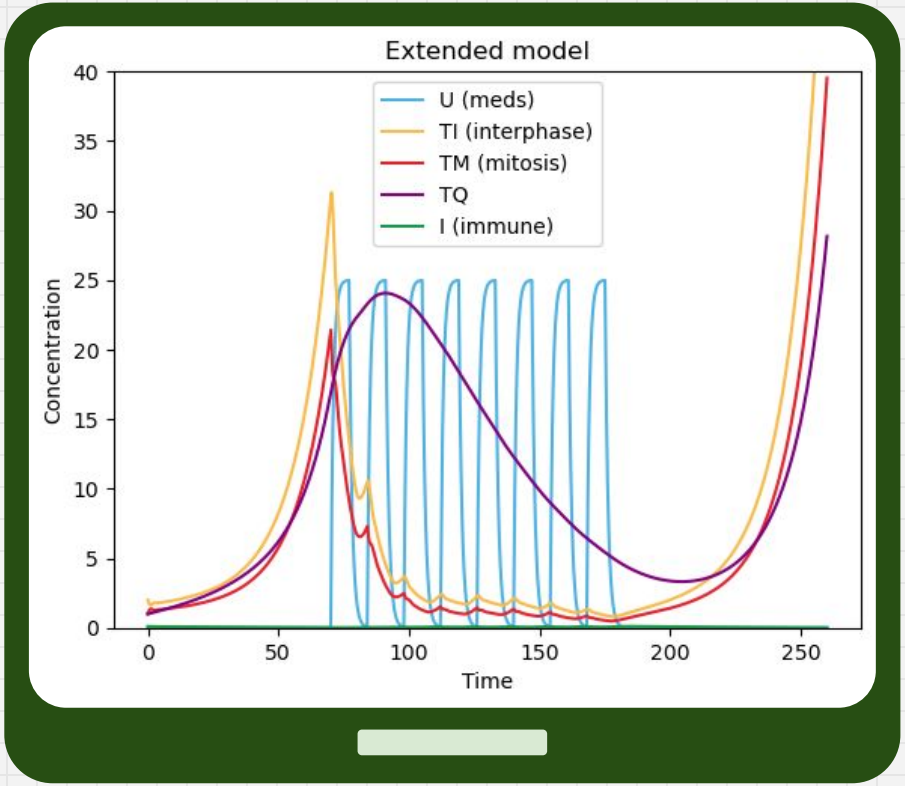
$$-a_q T_M$$

Assumptions & Modifications

- Mitosis is an instantaneous process.
- The drug is cytotoxic.
- The drug only arrests tumor cells in mitosis.
- **The Quiescent period G_0 is taken into consideration!**
- The concentration of the drug decays exponentially with time.

Impact of Drug Introduction to Cancer Cells without and with taking QCCs into consideration





Results

- Without taking QCCs into consideration, the drug interaction completely eradicates the proliferating cells.

- Taking QCCs into consideration, the same drug regimen does does not eradicate cancer, and the cancer relapses.

Works Cited

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Thank you.